



# Modelling Financial Markets

## Chapter 17: Bubbles and Crashes



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## Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EM: Expectation-Maximisation (algorithm)
- ▣ FRED: Federal Reserve Economic Data
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPL: Log-Periodic Power Law
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MC: Monte Carlo (simulation)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States

## Chapter Route and Outcomes

### Route of this chapter

- Question: prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value; can statistics detect such episodes **in real time**, and say **when** they end?
- Part I: what a bubble is, run-ups and crashes in the data, tests for explosive prices and their date-stamping
- Part II: the LPPLS model, regime switching, crypto and AI stocks, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#) at the end

### After this chapter you can

- separate the fundamental value from a rational bubble and state what can be tested
- test for explosive prices with GSADF and date-stamp episodes with BSADF, with valid critical values
- fit LPPLS, judge the stability of its critical time and evaluate its alarms honestly
- describe booms and busts with a Markov-switching model and test the number of regimes

## Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain;  
Wikimedia Commons

- **Tulips, 1636–1637:** contracts on rare bulbs traded at very high prices until the market stopped in February 1637; part of the path fits the fundamentals of a new luxury good [Garber \(1990\)](#)
  - ▶ Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720:** shares rose several-fold within months, then collapsed; a financial innovation, easy credit and a story about future profits [Kindleberger & Aliber \(2005\)](#)
  - ▶ Lesson 2: bubbles come with credit and with high turnover [Scheinkman & Xiong \(2003\)](#)
- **1929:** a boom partly financed with borrowed money ended in the October crash; Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic [Kindleberger & Aliber \(2005\)](#)
  - ▶ Lesson 3: we need tests that work on a single time series, in real time, before the panic

## Data Used in This Chapter

| Series                                                            | Frequency      | Period    | Use                      |
|-------------------------------------------------------------------|----------------|-----------|--------------------------|
| S&P Composite, real price and dividend                            | monthly        | 1871–2026 | price-dividend ratio     |
| 48 US industry portfolios                                         | monthly        | 1926–2026 | run-ups and crashes      |
| Nasdaq 100, S&P 500                                               | daily, monthly | 1990–2026 | dot-com, AI              |
| BET, BET-FI                                                       | daily, monthly | 1997–2026 | Bucharest Stock Exchange |
| Bitcoin, Ethereum, Dogecoin                                       | daily, weekly  | 2014–2026 | crypto booms             |
| NVIDIA, Cisco, Tesla, GameStop, Strategy (formerly MicroStrategy) | daily, monthly | 1990–2026 | single stocks            |

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: [shillerdata.com](https://shillerdata.com) for the S&P Composite; [Kenneth French Data Library](https://kennethfrench.com) for industries

## Fundamental Value

- No-arbitrage alone gives  $P_t = \mathbb{E}_t[M_{t+1}(P_{t+1} + D_{t+1})]$ , with  $M$  the stochastic discount factor and  $D$  the dividend
  - **Model assumption:** a constant required (expected) return  $r$ , so  $P_t = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{1 + r}$ ;  $\mathbb{E}_t$ : expectation under the physical measure, given the information at  $t$
- Solving forward gives the **fundamental value**:  $F_t = \sum_{i \geq 1} \frac{\mathbb{E}_t[D_{t+i}]}{(1 + r)^i}$ 
  - Gordon case, dividends growing at  $g < r$ :  $F_t = D_{t+1}/(r - g)$
- But  $P_t = F_t + B_t$  also solves the equation whenever  $\mathbb{E}_t[B_{t+1}] = (1 + r) B_t$
- $B_t$  is the **rational bubble**: investors pay for it because they expect to resell it at a higher price [Blanchard & Watson \(1982\)](#)

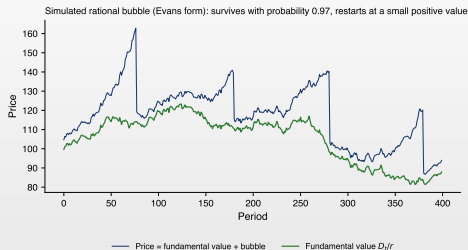
## Why a Rational Bubble Must Explode

- From  $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$ :  $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$ , an **explosive** path
  - ▶ The bubble grows faster than dividends ( $r > g$ ), so its share of the price rises over time
- A negative bubble would eventually push the expected price below zero, and a bubble that is zero today must stay zero: rational bubbles are positive and exist from the first trade [Diba & Grossman \(1988\)](#)
- A bubble is the failure of the **transversality condition**  $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$ : the price contains a value never paid out as dividends (derivation in the [Appendix](#))
- **Testable implication**: if dividends are at most  $I(1)$ , then without a bubble the price and the price-dividend ratio are **not explosive**
  - ▶  $I(1)$ : integrated of order one; the first difference is  $I(0)$  (stationary), while the level is not

## Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability  $\pi$  and collapses with probability  $1 - \pi$  [Blanchard & Watson \(1982\)](#)
  - ▶  $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$  with probability  $\pi$ ,  $B_{t+1} = \varepsilon_{t+1}$  otherwise
- Expected growth is still  $r$ ; while it survives the bubble grows faster, at  $(1+r)/\pi - 1$ , to pay for the crash risk
- Expected lifetime:  $1/(1-\pi)$  periods, a geometric waiting time
- After a collapse  $B = \varepsilon$  can be negative, which Diba–Grossman rule out: the simulation on the next slide restarts from a positive value (Evans form) [Evans \(1991\)](#)

## A Simulated Rational Bubble



- Fundamental value  $D_t/r$ , random-walk dividends;  $r = 1\%$ ,  $\pi = 0.97$ : growth close to 4.1% per period while alive and large, expected life 33.3 periods
- Evans form: collapse to  $\delta = 5$ , survival  $B_{t+1} = \delta + \frac{1+r}{\pi}(B_t - \frac{\delta}{1+r})$  times noise of mean 1, so  $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$  exactly
- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 1.43 times the fundamental value
- Between episodes the price tracks fundamentals, so a full-sample test sees mostly “normal” data: the tests of this chapter use **windows**

## Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017);

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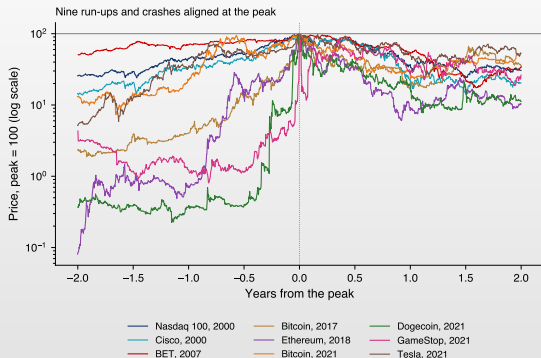
- **Feedback and narratives:** rising prices attract new buyers, “new era” stories spread by word of mouth [Shiller \(2015\)](#)
- **Disagreement and short-sale limits:** the price includes the option to resell to a more optimistic buyer [Scheinkman & Xiong \(2003\)](#)
- **Riding the bubble:** informed traders stay in as long as others do, since no one can burst it alone [Abreu & Brunnermeier \(2003\)](#), [Brunnermeier & Nagel \(2004\)](#)
- All three give prices that grow faster than fundamentals for a while: the explosive signature we will test

## Nine Episodes: Run-Up and Crash

| Episode          | Peak        | 2-year run-up | Max. fall | Trough   | Back at peak |
|------------------|-------------|---------------|-----------|----------|--------------|
| Nasdaq 100, 2000 | 27 Mar 2000 | +288%         | −83%      | Oct 2002 | Nov 2015     |
| Cisco, 2000      | 27 Mar 2000 | +592%         | −89%      | Oct 2002 | Aug 2021     |
| BET, 2007        | 24 Jul 2007 | +97%          | −83%      | Feb 2009 | Mar 2021     |
| Bitcoin, 2017    | 16 Dec 2017 | +4186%        | −83%      | Dec 2018 | Nov 2020     |
| Ethereum, 2018   | 13 Jan 2018 | +124004%      | −94%      | Dec 2018 | Feb 2021     |
| Bitcoin, 2021    | 8 Nov 2021  | +667%         | −77%      | Nov 2022 | Mar 2024     |
| Dogecoin, 2021   | 7 May 2021  | +26897%       | −92%      | Jun 2022 | not yet      |
| GameStop, 2021   | 27 Jan 2021 | +2218%        | −88%      | Apr 2024 | not yet      |
| Tesla, 2021      | 4 Nov 2021  | +1837%        | −74%      | Jan 2023 | Dec 2024     |

- Peak: highest daily close in the episode; run-up: price change over the two years before the peak; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

## Run-Ups and Crashes Aligned at the Peak



- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ enormously, from +97% (BET) to +124004% (Ethereum); the falls are much more alike, 74–94%
- Common shape: an accelerating rise (convex on the log scale, the growth rate itself rises), a sharp top, a deep fall: the **explosive** behaviour the tests look for
- Selection bias: we picked these episodes because they crashed; many run-ups did not

## The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia

[Commons](#)

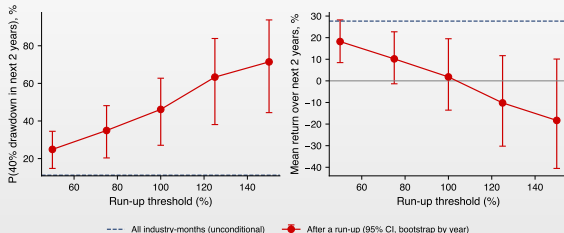
- After the peak of 27 Mar 2000: Nasdaq 100  $-83\%$ , Cisco  $-89\%$  (table); Cisco regained its peak only in Aug 2021
- Hedge funds rode the bubble and cut positions before the fall [Brunnermeier & Nagel \(2004\)](#)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble [Pástor & Veronesi \(2009\)](#)

## Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news [Fama \(2014\)](#)
- Greenwood, Shleifer and You answer with **all** US industries since 1926, not with famous episodes only: do sharp run-ups predict crashes, or low returns? [Greenwood, Shleifer & You \(2019\)](#)
- **Run-up**: value-weighted industry return over the past 2 years above a threshold (e.g. 100%), both raw and net of the market
  - ▶ And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted (Section 2 of the paper)
  - ▶ First month of the run-up; a new episode in the same industry only two years later
- **Crash**: a drawdown of at least 40% from a previous high within the next 2 years
- Our replication, same sample rules: the first 48 of the 49 industries of the [Kenneth French Data Library](#) (without "Other"), months with at least ten firms: 41 679 industry-months, 1926–2026

## What Do Sharp Run-Ups Predict?

Bubbles for Fama, 48 US industries 1926-2026: two-year run-ups above the threshold, raw and net of the market

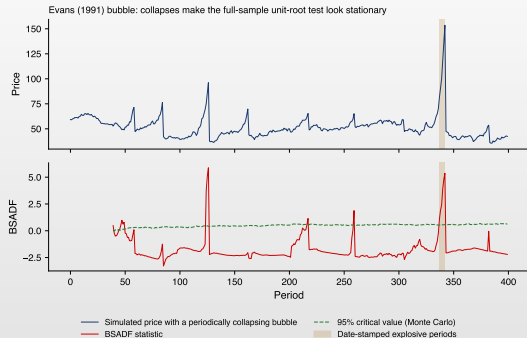


- After a run-up above 100% (52 episodes in 27 industries), a 40% crash followed in 46% of cases, 95% CI [27%, 63%], against 11% for a random industry-month; above 150%: 71% ( $n = 21$ )
  - CI: bootstrap over calendar years, because run-ups cluster in time (e.g. 1928–1929, 1967–1968, 1980); effective sample size in the [Appendix](#)
- The mean return over the next 2 years is not significantly different from zero: 2% after a 100% run-up, CI [–14%, 20%]: no statistically established negative mean return
- Run-ups raise the **risk** of a crash, not the predictable loss: price run-ups alone do not give a profitable timing signal Greenwood, Shleifer & You (2019)

## Right-Tailed ADF and Collapsing Bubbles

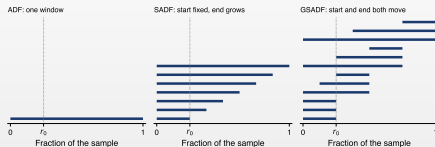
- Regression on the log price  $y_t$ :  $\Delta y_t = a + b y_{t-1} + \varepsilon_t$ ;  $ADF = \hat{b}/SE(\hat{b})$ ; here with a constant and no lagged  $\Delta y$
- Bubble test:  $H_0 : b = 0$  (unit root) vs  $H_1 : b > 0$  (explosive root), the **right tail**; the left-tail Dickey–Fuller table does not apply [Phillips, Wu & Yu \(2011\)](#)
- Evans: a bubble that collapses periodically makes the **full-sample** series look stationary or like a random walk: low power; collapses give the full-sample slope a negative sign ( [Appendix](#) ) [Evans \(1991\)](#)
- Solution: run the ADF regression on many **windows** and take the largest statistic
  - ▶ SADF (Phillips, Wu and Yu): windows start at the first observation and grow [Phillips, Wu & Yu \(2011\)](#)
  - ▶ GSADF (Phillips, Shi and Yu): windows may start and end anywhere [Phillips, Shi & Yu \(2015a\)](#)

## Evans Bubble: Full Sample vs Windows



- 400 simulated periods; BSADF (defined below) against its 95% critical value from 2000 random-walk paths
- Full-sample right-tailed ADF:  $-5.49$  vs 95% critical value  $-0.09$ : no rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46) and GSADF 5.88 (critical value 2.21)
- BSADF crosses its critical value in several bursts, but only one lasts long enough to be date-stamped: the minimum-duration rule trades short bursts for fewer false alarms

## Three Ways to Choose Windows: SADP, GSADF, BSADF

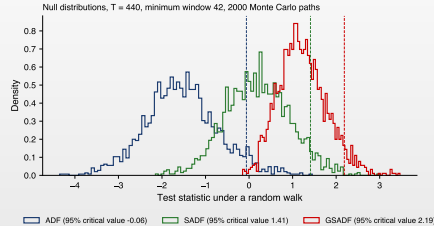


- $ADF_{r_1}^2$ : the ADF statistic on the window from fraction  $r_1$  to fraction  $r_2$  of the sample; each bar in the chart is one ADF regression
- $SADF(r_0) = \sup_{r_2 \in [r_0, 1]} ADF_0^2$  Phillips, Wu & Yu (2011)
- $GSADF(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^2$  Phillips, Shi & Yu (2015a)
- Backward SADF at the end point  $r_2$ :  $BSADF_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^2$ ;  $GSADF = \sup_{r_2} BSADF_{r_2}$ 
  - $BSADF_{r_2}$  uses only data up to  $r_2$ : it can be computed **in real time**
- Minimum window rule:  $r_0 = 0.01 + 1.8/\sqrt{T}$ , e.g.  $T = 440$  months:  $r_0 = 0.096$ , 42 months Phillips, Shi & Yu (2015a)

## Date-Stamping: When Did It Start and End?

- **Origination:** the first date at which BSADF exceeds its critical value [Phillips, Shi & Yu \(2015a\)](#)
  - ▶ The critical value sequence  $cv_{r_2}$ : 95% quantile of  $BSADF_{r_2}$  under the null, date by date
- **Termination** (PSY, eq. 8): the first date after  $\hat{r}_e + \delta \ln(T)/T$  at which BSADF falls below its critical value,  $\delta$  depending on the frequency [Phillips, Shi & Yu \(2015a\)](#)
- **Rule used in this chapter:** the minimum-duration rule of Vasilopoulos, Pavlidis and Martínez-García (R package exuber): runs of consecutive exceedances shorter than  $L$  are dropped [Vasilopoulos, Pavlidis & Martínez-García \(2022\)](#)
  - ▶  $L = \lceil \ln T \rceil$  ( $\delta = 1$ );  $T = 440$ : 7 months. Start: first exceedance; end: last exceedance (the package dates the end one period later). Eq. 8 would instead extend a one-date blip to  $L$  dates
  - ▶ Real time: the start is dated in hindsight; the episode is **confirmed** only after  $L$  consecutive exceedances,  $L - 1$  periods later
- PWY date-stamping uses the recursive  $ADF_0^2$  instead: it may miss a second bubble after a collapse [Phillips, Shi & Yu \(2015b\)](#)

## Null Distributions and Critical Values



- Random walk with a small drift,  $y_t = T^{-1} + y_{t-1} + \varepsilon_t$ , 2000 paths of  $T = 440$ ; 95% critical values: a single ADF  $-0.06$  (negative!), SADF 1.41, GSADF 2.19 [Phillips, Shi & Yu \(2015a\)](#)
- A supremum over more windows shifts the distribution to the right: the GSADF bar is the price of searching over about 80 000 windows; simulate it for each  $T$ ,  $r_0$  and lag order
- Asymptotics ([Appendix](#)): the null limit is pivotal; against mildly explosive roots the tests are consistent; date-stamping keeps a 5% false-alarm chance at every date [Phillips, Shi & Yu \(2015b\)](#)

## Changing Volatility: Wild Bootstrap

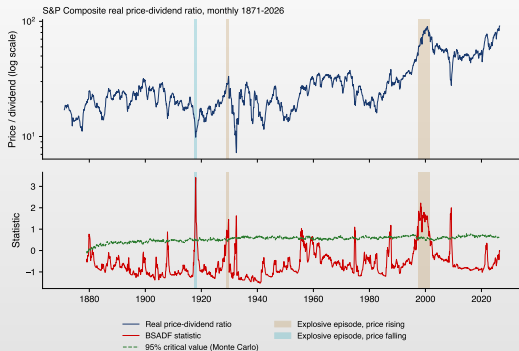
- ▣ Monte Carlo critical values assume shocks with constant variance; real prices have volatility clusters and shifts
- ▣ A burst of volatility can mimic explosiveness and inflate the rejection rate [Harvey et al. \(2016\)](#)
- ▣ **Wild bootstrap**: multiply the observed, demeaned changes  $\Delta y_t$  by random signs  $w_t = \pm 1$ , rebuild the path, recompute BSADF [Phillips & Shi \(2020\)](#)
  - ▶ The bootstrap null keeps the volatility pattern of the data but has no explosive root
  - ▶ Centred with the full-sample mean and resampling the whole path: **retrospective** critical values; in real time, recompute them with data up to each date
- ▣ Validity: the bootstrap data have a unit root by construction and inherit the time pattern of  $\text{Var}(\Delta y_t)$ ; demeaning removes the drift, as in the null model [Harvey et al. \(2016\)](#)

## Size of GSADF When Volatility Changes: Monte Carlo

| Null model, no bubble                        | GSADF, MC crit. value | GSADF, wild crit. value | False episode, MC | False episode, w |
|----------------------------------------------|-----------------------|-------------------------|-------------------|------------------|
| Constant volatility                          | 4.1%                  | 3.9%                    | 49.1%             | 51.9%            |
| Volatility $\times 3$ at $T/2$               | 32.0%                 | 4.1%                    | 63.8%             | 49.6%            |
| Volatility $\times 1/3$ at $T/2$             | 4.3%                  | 6.1%                    | 48.0%             | 54.8%            |
| GARCH(1,1), $\alpha = 0.10$ , $\beta = 0.88$ | 16.6%                 | 6.8%                    | 58.1%             | 52.8%            |

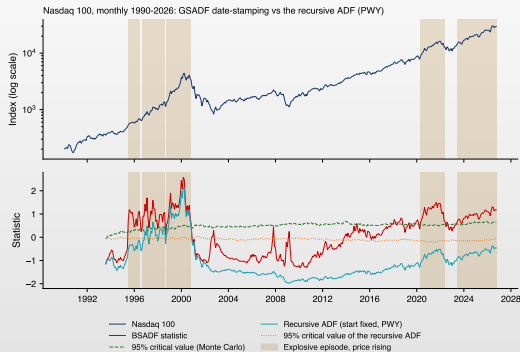
- Rejection rates at 5%;  $y_t = T^{-1} + y_{t-1} + e_t$ ,  $T = 400$ , minimum window  $w_0 = \lfloor r_0 T \rfloor = 40$  observations, 1 000 paths per row, each with its own wild bootstrap (199 replications); Monte Carlo error  $\pm 1.4$  points near 5%; false episode: BSADF above its pointwise 95% value for 6 periods
- A late rise in volatility is mistaken for explosiveness (32.0% with Monte Carlo, 4.1% with wild critical values); under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless [Harvey et al. \(2016\)](#)
- Even with constant volatility 49.1% of the paths contain a false date-stamped episode: the size is controlled date by date, not over the sample

## S&P 500 Price-Dividend Ratio, 1871–2026



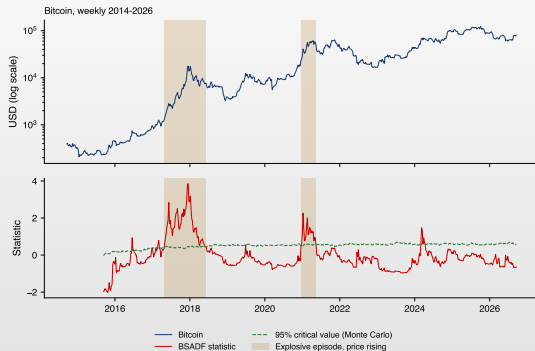
- Real price over real dividend, monthly, Jan 1871–Jun 2026,  $T = 1865$ , minimum window 96 months
- GSADF = 3.41 vs 95% critical value 2.39 (99%: 2.88); the full-sample ADF ( $-1.11$ ) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
  - ▶ 1917–1918 is an accelerating **fall** of the ratio (real prices fell faster than real dividends): explosive, but not a bubble
- Dot-com: peak 90.2 in Aug 2000 vs a long-run mean of 29.8; the sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not an explosive one
- A ratio can explode through its denominator: always check both price and dividend

## Nasdaq 100: GSADF vs PWY Date-Stamping



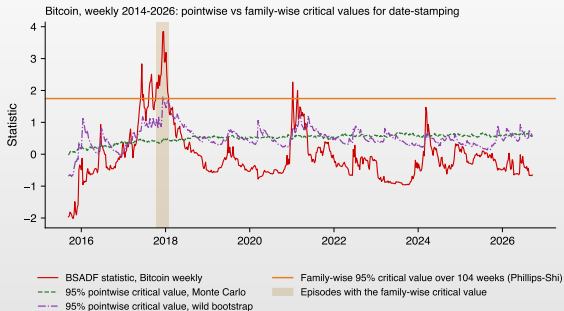
- Monthly log closes 1990–2026,  $T = 440$ , minimum window 42 months; 2 000 Monte Carlo paths
- GSADF = 2.57 (95% critical value 2.19); SADF = 2.14 (critical value 1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
  - ▶ PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF starts in 1990 and is dragged down by the long flat period after 2000: it misses the post-2020 episodes that BSADF finds
- Dot-com episode dated from Jun 1995, 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995

## Bitcoin: Explosive Episodes



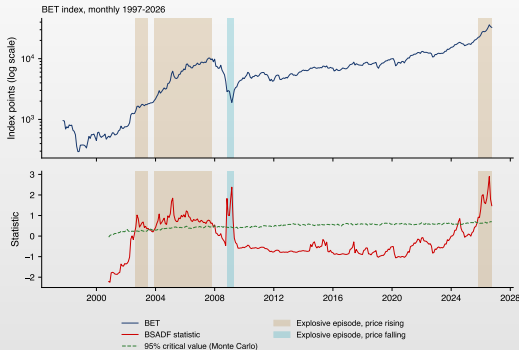
- Weekly log closes 2014–2026,  $T = 626$ , minimum window 51 weeks; Monte Carlo critical values
- GSADF = 3.85: Monte Carlo 95% critical value 2.27, wild bootstrap 2.82: rejected either way
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
  - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: episode dated from Apr 2017, 8 months before the peak of 16 Dec 2017, confirmed in real time in Jun 2017; it ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged: steady growth is not the same as explosive acceleration

## Date-Stamping Is a Multiple Test: Bitcoin



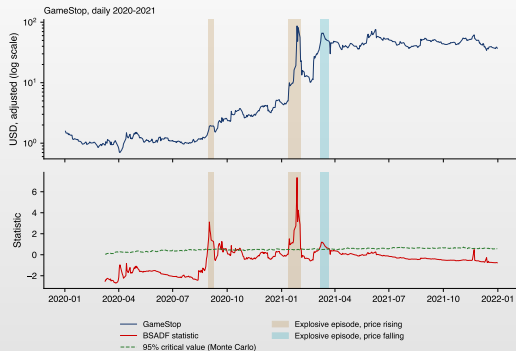
- BSADF is compared with its 95% critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER): the probability of at least one false alarm over the whole monitoring window; in 2 000 random walks with the Bitcoin  $T$  and  $w_0$ : a crossing in 95%, an “episode” in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF over the last  $\tau_b$  monitoring dates (here  $\tau_b = 104$  weeks); family-wise critical value 1.75, against pointwise values of at most 0.71 [Phillips & Shi \(2020\)](#)
- Episodes above the higher threshold: Oct 2017–Jan 2018; the 2020–2021 episode found with pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the [Appendix](#))

## BET: The 2007 Boom and 2025



- Monthly log closes of the official BET, 1997–2026,  $T = 348$ , minimum window 37 months
- $GSADF = 2.90$  vs 95% critical value 2.21; episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up before and around Romania's accession to the European Union (January 2007); BET then fell 83% from its peak of 24 Jul 2007
- Oct 2008–Mar 2009 is an accelerating **fall**: explosive can mean down as well as up
- The current episode (from Oct 2025) is still open: explosive does not mean it will crash; it means the rise is accelerating

## GameStop, January 2021



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- Daily log closes 2020–2021,  $T = 504$ , window 45 days; GSADF = 7.34, critical value 2.21
- January episode dated from 13 Jan 2021, 14 days before the closing high of 27 Jan 2021; confirmed on 22 Jan 2021
- A surge of weeks: it needs daily data; the test does not identify the mechanism (short covering)

 MFM\_ch17\_bet\_gamestop

## Real Time vs Hindsight

| Episode              | Start / Peak              | Confirmed   | Lead (m.) | Lead, conf. | Rise after conf. | Fall after peak |
|----------------------|---------------------------|-------------|-----------|-------------|------------------|-----------------|
| Nasdaq 100, monthly  | Jun 1995 / Mar 2000       | Dec 1995    | 57        | 51          | +716%            | −76%            |
| BET, monthly         | Nov 2003 / Jul 2007       | Apr 2004    | 44        | 39          | +299%            | −83%            |
| Bitcoin 2017, weekly | Apr 2017 / Dec 2017       | Jun 2017    | 8         | 6           | +590%            | −83%            |
| Bitcoin 2021, weekly | Dec 2020 / Apr 2021       | Feb 2021    | 4         | 2           | +66%             | −75%            |
| GameStop, daily      | 13 Jan 2021 / 27 Jan 2021 | 22 Jan 2021 | 0.5       | 0.2         | +435%            | −88%            |

- Start: first exceedance of the chain of BSADF episodes that leads to the peak (hindsight); confirmed: after  $L$  consecutive exceedances, the first date known in real time; leads in months; rise: confirmation to peak; fall: to the lowest close within 2 years (definitions in the [Appendix](#))
- Even at confirmation the tests **warn early**, but the lead ranges from 5 days to about 4.2 years, and selling at confirmation would have missed most of the rise: they do not time the crash [Phillips, Shi & Yu \(2015b\)](#)

## Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0  
(Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS): imitation among traders makes the **hazard rate** of a crash rise as a critical time  $t_c$  approaches [Johansen, Ledoit & Sornette \(2000\)](#)
- Rational investors stay in only if the price grows fast enough to pay for the rising crash risk
- Result: **faster-than-exponential** growth, decorated by oscillations that speed up near  $t_c$
- $t_c$  is the most probable end of the bubble, not a certain crash date [Sornette \(2017\)](#)

## The LPPLS Equation

- $\ln p(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
- $A$ : log price at  $t_c$ ;  $B < 0$ : the super-exponential trend for a rising bubble;  $0 < m < 1$ : power-law exponent
  - ▶ The growth rate of the power-law trend,  $-Bm(t_c - t)^{m-1}$ , rises without bound as  $t \rightarrow t_c$
- $C, \omega, \phi$ : amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters:  $A, B, C, \phi$  and the nonlinear  $t_c, m, \omega$
- LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at  $t_c$

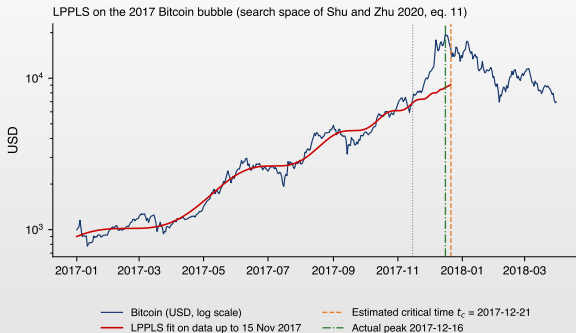
## Calibration: Filimonov and Sornette

- Rewrite the cosine:  $C \cos(\omega \ln(t_c - t) - \phi) = C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))$  [Filimonov & Sornette \(2013\)](#)
  - ▶  $A, B, C_1, C_2$  are now **linear**: for given  $(t_c, m, \omega)$  they come from OLS
- Search only over  $(t_c, m, \omega)$ : a grid, then a Nelder–Mead refinement of the sum of squared residuals
- Search space of Shu and Zhu (Section 2.3, eq. 11):  $m \in [0, 1]$ ,  $\omega \in [1, 50]$ ,  $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$ , damping  $\frac{m|B|}{\omega|C|} \geq 1$ , an approximate non-negativity condition for the crash hazard rate, the crash probability per unit time given no crash yet (exact:  $m|B| \geq |C|\sqrt{m^2 + \omega^2}$ , Seminar 17, A8) [Shu & Zhu \(2020\)](#)
- Many local minima remain: different windows and starting points give different  $t_c$  [Geraskin & Fantazzini \(2013\)](#)

## Qualified Fits: The Filter

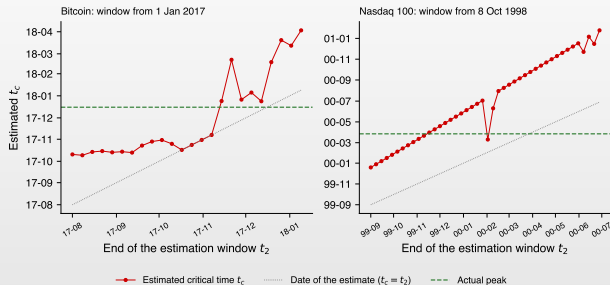
- A fit on  $[t_1, t_2]$  is **qualified** if it passes all conditions of Shu and Zhu (Section 2.3, eq. 12), and  $B < 0$  for a rising bubble Shu & Zhu (2020)
  - ▶  $m \in [0.01, 0.99]$ ,  $\omega \in [2, 25]$ ,  $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
  - ▶ Number of oscillations (half-periods)  $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$ , so the oscillations are not noise
  - ▶ Largest relative error of the fitted price  $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
  - ▶ Lomb periodogram of the detrended residual  $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$ : log-periodicity significant at 10%
  - ▶ Residual unit-root check for  $\ln \hat{p}_t - \ln p_t$ : Dickey–Fuller and Phillips–Perron reject a unit root at 10% (a stationary, mean-reverting residual; this alone does not make it AR(1))
- $[t_1, t_2]$ : the estimation window; the filter is applied to each window separately

## LPPLS on Bitcoin, 2017



- Fit on daily log prices from 1 January to 15 November 2017; the path after 15 November is out of sample
- The fit follows the 2017 rise and its oscillations closely:  $m = 0.95$ ,  $\omega = 10.54$ ,  $B = -2.62$
- Estimated  $t_c$ : 21 Dec 2017; actual peak: 16 Dec 2017, i.e. 5 days earlier
- The fit fails the condition on relative error  $\leq 0.15$ : it does **not** pass the filter, even though it looks convincing

## How Stable Is the Critical Time?

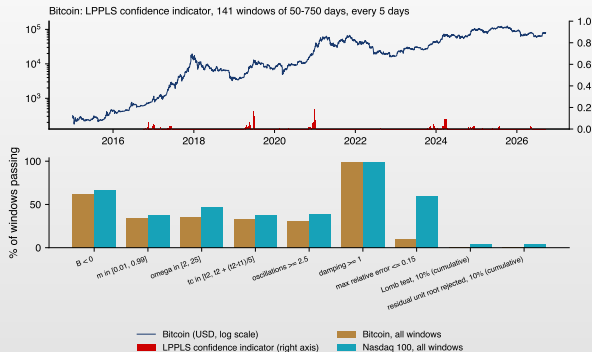


- A good in-sample fit is not a forecast: each point is one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits)  $t_c$  ranged from Oct 2017 to Feb 2018; mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits)  $t_c$  ranged from Dec 1999 to Sep 2000; mean absolute error 70 days;  $t_c \geq t_2$  is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin) and 0% (Nasdaq 100); the birth of a bubble is easier to diagnose than its burst [Demos & Sornette \(2017\)](#)

## The LPPLS Confidence Indicator

- At each date  $t_2$ , fit many windows  $[t_2 - L, t_2]$  and report the **share of qualified fits** [Demirer et al. \(2019\)](#)
  - ▶ Grid of Shu and Zhu (Section 2.3): 141 windows,  $L$  from 750 down to 50 observations in steps of 5;  $t_2$  every 5 observations; data up to  $t_2$  only [Shu & Zhu \(2020\)](#)
  - ▶ Daily closes; Bitcoin trades every day, so one observation is one calendar day
- A high share means the bubble signature is robust to the choice of window
- Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
  - ▶ Fall measured from the price on the signal date; only dates whose 90 calendar days are fully observed (the last three months of data are dropped)

## Bitcoin: Real-Time Confidence Indicator



- Top: share of the 141 windows with a qualified fit, every 5 days, 2016–2026; bottom: share of windows passing each condition (Lomb and residual tests cumulative)
- Bitcoin: the indicator was positive on 14% of the 2017 dates, 3% in 2021 and 12% in other years; maximum 0.18 in Dec 2020
- After a signal, a 20% fall within 90 days followed in 22% of cases, vs 37% without a signal (709 dates)
  - Nasdaq 100, 1997–2026: 4% after a signal vs 11% without (1482 dates)
- The raw difference is not yet evidence: signals come in runs and the 90-day outcome windows overlap (test on the next slide)

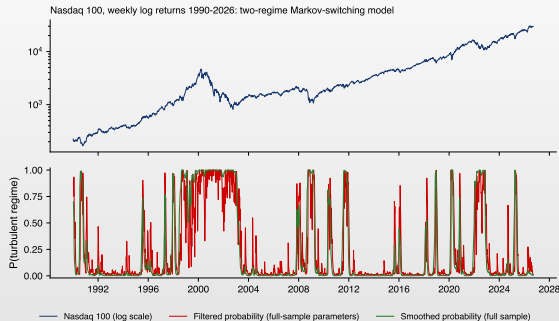
## Is the Indicator Better than Chance? A Test

- $H_0: P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$ ; statistic  $\Delta = \hat{\rho}_{\text{on}} - \hat{\rho}_{\text{off}}$
- Dates 5 days apart share up to 85 of their 90 outcome days: the two-sample z-test, which assumes independent dates, understates the standard error
- Circular block bootstrap of the (signal, outcome) pairs, blocks of 90 days, 9 999 replications; linear probability regression with Newey–West standard errors
  - ▶ Bitcoin:  $\Delta = -15.5$  points, 95% CI  $[-30.0, 1.8]$ ; block SE 8.1 vs independent-data SE 5.0; regression  $p = 0.049$
  - ▶ The same test for the Nasdaq 100, with a power calculation: Seminar 17, Exercise B6
- The block-bootstrap CI contains zero, while the Newey–West regression is borderline ( $p = 0.049$ ): weak evidence, and the point estimate is negative, i.e. if anything a lower crash rate after signals; a signal is **not** followed by a large fall more often than no signal, because signals come in steady rises, not just before crashes

## The Markov-Switching Model

- Weekly log return  $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$ ,  $\varepsilon_t \sim N(0, 1)$ , regime  $s_t \in \{1, 2\}$  unobserved [Hamilton \(1989\)](#)
  - ▶ Each regime has its own mean  $\mu$  and volatility  $\sigma$ : a calm and a turbulent regime
- Regimes follow a Markov chain:  $p_{ij} = P(s_t = j \mid s_{t-1} = i)$ ; expected duration of regime  $i$ :  $1/(1 - p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates  $P(s_t = j \mid r_1, \dots, r_t)$  week by week
- **Filtered** probability: returns up to  $t$  for given parameters; **smoothed**: the whole sample (hindsight)
  - ▶ Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to  $t$
  - ▶ A worked filter step is in Seminar 17, Exercise A5

## Nasdaq 100: Two Regimes

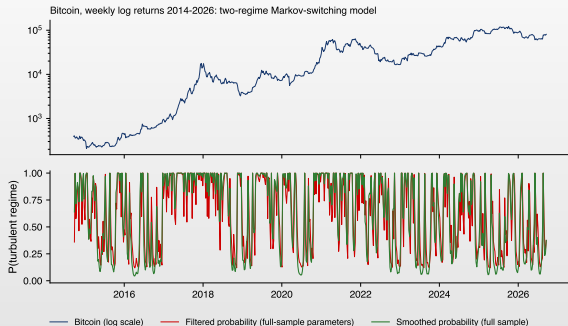


| Regime    | $\hat{\mu}$ (SE) | $\hat{\sigma}$ | $\hat{\rho}_{ii}$ | Dur. |
|-----------|------------------|----------------|-------------------|------|
| Calm      | 0.43 (0.06)      | 2.30           | 0.987             | 75.4 |
| Turbulent | -0.32 (0.27)     | 5.40           | 0.957             | 23.5 |

- Weekly log returns in %, 1990–2026, 1 915 weeks; duration in weeks; calm regime 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5); filtered probability 0.99 at the end of March 2000: prices kept rising while volatility jumped

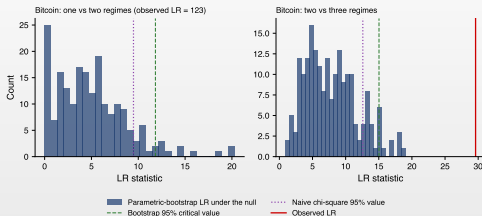
 MFM\_ch17\_markov\_switching

## Bitcoin: Two Regimes



- Weekly log returns 2014–2026, 626 weeks
- Calm:  $\hat{\mu} = 0.26\%$  per week,  $\hat{\sigma} = 2.93\%$ , duration 3.8 weeks
  - ▶ Turbulent:  $\hat{\mu} = 1.20\%$  (SE 0.58),  $\hat{\sigma} = 11.22\%$ , duration 6.3 weeks, 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean; regimes are short: volatility states, not bubble phases
- Regimes ignore the price level (a slow, calm bubble stays calm): BSADF says “the price is accelerating”, the regime says “risk has changed”

## How Many Regimes? A Bootstrap LR Test

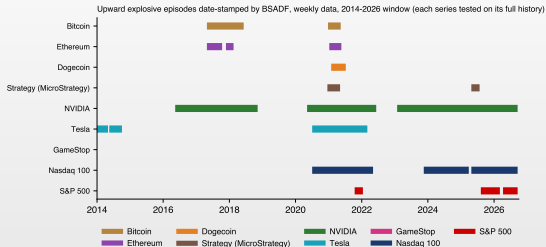


- $H_0$ : one regime (iid Normal returns) vs  $H_1$ : two Markov regimes,  $LR = 2(\hat{\ell}_2 - \hat{\ell}_1)$ ; not  $\chi^2$ :  $p_{11}, p_{22}$  are not identified under  $H_0$  ([Appendix](#)) [Hansen \(1992\)](#)
- **Parametric bootstrap**: simulate from the fitted  $H_0$  model, refit both models, recompute LR; 199 replications;  
 $p = (1 + \#\{LR^* \geq LR\}) / (B + 1)$
- One vs two: Bitcoin LR = 123.2 (95% value 11.8,  $p = 0.005$ ); Nasdaq 100 LR = 475.7 (11.1,  $p = 0.005$ )
- Two vs three, Bitcoin: LR = 29.6, bootstrap 95% value 15.0 vs naive  $\chi^2_6$  value 12.6;  $p = 0.005$  (Seminar 17, B8)
- One regime is rejected for both series; a third Bitcoin regime is also supported. Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes

## Are Crypto-Assets Bubbles?

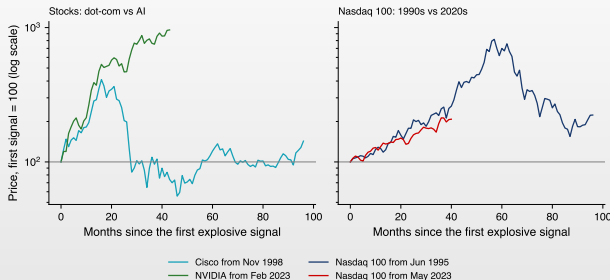
- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero [Cheah & Fry \(2015\)](#)
- Alternative: value from network adoption (Metcalf's law), with LPPL bubbles on top [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble [Pele et al. \(2023\)](#)
- Without a fundamental value we can still test the price: is it explosive, and when?

## Explosive Episodes in Crypto and Tech Stocks



- Weekly log closes; each series tested on its own history with its own Monte Carlo critical values; rising episodes only
- Crypto: Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021: booms cluster in 2017 and late 2020–2021
- Strategy (formerly MicroStrategy; holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
- GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration; an accelerating **fall** also gives  $b > 0$ : always report the direction

## Dot-Com vs AI



- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
- Cisco: first signal Nov 1998, peak after 16 months at 4.1 times the signal price; NVIDIA: first signal Feb 2023, 43 months later at 9.6 times, episode still open
- Nasdaq 100 from Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months, a slower rise than in the 1990s

## Is There an AI Bubble? What the Tests Can Say

- Monthly BSADF flags NVIDIA (Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing) and the Nasdaq 100 (current episode from May 2023)
- The tests cannot separate a bubble from explosive fundamentals: fast-growing expected profits of a new technology can also make prices explosive [Pástor & Veronesi \(2009\)](#)
- History: Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- What we can say: the tests detect explosive price dynamics; the industry run-up evidence of Greenwood, Shleifer and You (2019, *Bubbles for Fama*) motivates watching crash risk but does not quantify it for these stocks, and the timing is unknown

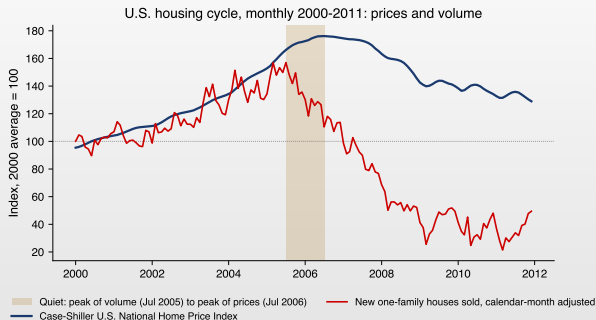
## Case Study (1/5): the Research Question

- Paper: [DeFusco, Nathanson & Zwick \(2022\)](#), *Journal of Financial Economics*, 146(1), 205–229
  - ▶ Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Question: who buys during a boom, and can speculation explain the joint path of prices, transaction volume and unsold listings (Introduction)?
  - ▶ Speculation, in the Harrison–Kreps definition the authors adopt (footnote 1): paying more for an asset because one has the right to resell it
- Three phases (Section 2 of the paper): the boom (prices and volume rise), the “quiet” (volume falls while prices still rise and unsold listings pile up), the bust
- Why it matters: the explosiveness tests of this chapter use prices only; Greenwood, Shleifer and You (2019, *Bubbles for Fama*) [Greenwood, Shleifer & You \(2019\)](#) also study turnover, issuance and volatility, and bubbles come with high turnover [Scheinkman & Xiong \(2003\)](#)
  - ▶ Quantities show who drives a boom, as blockchain flows did for the 2017 Bitcoin boom [Griffin & Shams \(2020\)](#)

## Case Study (2/5): Data, Method and Headline Results

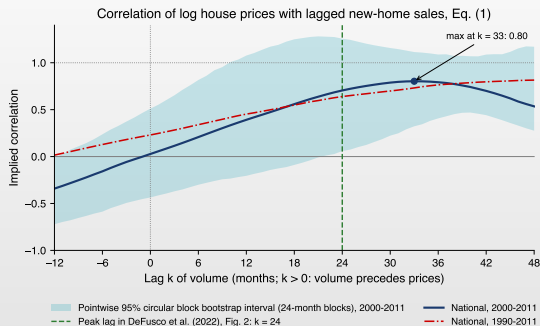
- Data (Section 1.1): 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas; **speculators**: short-term buyers who resell within three years, and non-occupant buyers (Section 1.2)
- Lead-lag (Eq. 1):  $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$ ,  $k = -12, \dots, 48$  months,  $p$  the demeaned log price,  $v$  the demeaned sales rate; implied correlation  $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$
- Figure 2: the correlation peaks at a lag of 24 months; Figure 4: short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: one standard deviation more speculative volume goes with a price boom larger by 24.9 points and a bust deeper by 7.4 points: cities with more speculation had larger booms and larger busts
- Model (Sections 5–6, Table 6): extrapolating buyers with short horizons amplify the cycle; a 5% purchase tax cuts the bust from 8.2% to 1.1%

## Case Study (3/5): Prices and Volume, 2000–2011



- FRED: Case-Shiller U.S. National Home Price Index and new one-family houses sold, 2000 average = 100; volume adjusted by calendar-month means, as in Figure 1 of [DeFusco, Nathanson & Zwick \(2022\)](#)
- Boom: sales peak at 157 in Jul 2005 (paper: 150% of the 2000 level); prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak, when sales are already down to 110 (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices; new-home sales exclude resales, so they cannot show who trades

## Case Study (4/5): Volume Leads Prices



- Eq. (1) on the national series, without a constant; pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- 2000–2011: maximum at  $k = 33$  (0.80, CI [0.32, 1.11]); at the paper's  $k = 24$  ours is 0.70, CI [0.06, 1.26]
- The band excludes zero for  $k = 22, \dots, 48$ , but one national cycle holds few independent blocks: wide intervals
- Direction as in the paper (volume first, by two to three years); its length is not precise (1990–2011:  $k = 48$ )

## Case Study (5/5): Limits and an Open Question

- Data limit of our illustration: the public series are national and cover new homes only, so short-term and non-occupant volume cannot be built; Figures 1–2 can be illustrated, Table 2 cannot (Seminar 17, C3)
- Limits stated by the authors
  - ▶ Holding periods are realised, not intended (footnote 4; instrumental-variable check in the Online Appendix)
  - ▶ The non-occupant proxy misclassifies owners who give the property address for tax mail (Section 1.2)
  - ▶ The counterfactuals may overstate the role of short-term buyers, because selling horizons are exogenous in the model; credit and new construction are left out (Sections 4.3, 6.5.3, 7)
- Open question: does trading volume also lead prices in crypto booms, and is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

## An Open Question

Can bubble alarms be raised in real time with a controlled false-alarm rate, and do they carry crash information?

- BSADF date-stamping runs one right-tailed test per date: the size is controlled date by date, not over the whole monitoring period
  - ▶ Live research: monitoring rules for bubbles *and* crashes with controlled false alarms Whitehouse, Harvey & Leybourne (2023); real-time “econometric thermometers” for housing Shi & Phillips (2023)
- Formal version, for an alarm  $A_t$  and a crash  $C_{t,h}$  (a fall of at least 20% within  $h = 90$  days):  
 $H_0: P(C_{t,h} | A_t = 1) = P(C_{t,h} | A_t = 0)$ ,  $H_1$ : larger after an alarm
  - ▶ Falsified if the hit rate after alarms does not exceed the rate without alarms, under inference that respects overlapping windows
- Identification: explosive prices need not be a bubble
  - ▶ Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009); run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

## The Discovery Loop and Its Checks

- **Literature:** ask for papers on the false-alarm rate of sequential bubble monitoring; resolve every DOI on [Crossref](#) and find the table the answer quotes
- **Hypotheses:** rational bubble, explosive expected cash flows, falling discount rate, each with its own observable implication (e.g. an explosive price-to-earnings ratio vs explosive earnings)
- **Code:** replicate a known number first ( $GSADF = 3.85$  for weekly Bitcoin), then extend
- **Robustness:** pre-register the grid of horizons and crash thresholds with a Holm correction; **critique:** ask the assistant to attack the result as a referee
- **Checks** the researcher must run (details and prompts in the [Appendix](#))
  - ▶ No full-sample information in a date stamp; right-tail critical values; wild bootstrap under changing volatility; multiplicity over dates and assets; inference robust to overlapping outcome windows

## Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal: “a positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher’s exact test”; alarms come in runs and outcomes overlap, so we test by circular rotations
- Bitcoin (709 dates, every 5 days): a fall of 20% within 90 days after 22% of alarms vs 37% without
  - ▶ Difference –15.5 percentage points; one-sided  $p = 0.96$  for  $H_1$  from 658 rotations of the alarm series
  - ▶ Nasdaq 100: 4% vs 11%, difference –7.1 points,  $p = 0.98$
- Pre-registered grid of 4 horizons  $\times$  3 thresholds: smallest raw  $p$  is 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
  - ▶ No cell rejects  $H_0$  at 5%, before or after the Holm correction
- Interpretation: the indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal (assumptions in the [Appendix](#))

## Your Turn: a Project Seed

- **Question:** can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period and still warn before large falls?
- **Replicate first:** GSADF = 3.85 for weekly Bitcoin, then the crash probability after run-ups above 100% in [Greenwood, Shleifer & You \(2019\)](#) (ours: 46%)
  - ▶ Extend: calibrate the alarm threshold by simulation over the whole monitoring period [Whitehouse, Harvey & Leybourne \(2023\)](#); evaluate the alarms with rotations or a block bootstrap
- **Deliverables:** a pre-registered grid fixed in a dated document before the evaluation, a notebook that regenerates every number, false-alarm rates and hit rates with valid  $p$ -values
- **AI log:** prompts that shaped the design in AI\_USE.md; each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it in AI\_ERRORS.md ([templates](#))

## Key Takeaways

- ▣ A bubble is price minus fundamental value; rational bubbles must be explosive, and bursting bubbles hide from full-sample tests
- ▣ GSADF tests for explosive episodes, BSADF date-stamps them in real time; with a late volatility rise Monte Carlo critical values reject 32.0% of true nulls, the wild bootstrap 4.1%; date-stamping needs family-wise control
- ▣ Across our episodes the tests warned months before the peak, but did not time the crash
- ▣ LPPLS describes the acceleration; its critical time moves with the window, and the confidence indicator is judged by a dependence-robust test of hit rates
- ▣ Sharp run-ups raise crash risk without a statistically established average loss: a risk-management signal, not a trading rule

## Which Tool When?

| Question                                                            | Tool                                                                                                                                                     |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Has there been any explosive episode?<br>When did it start and end? | GSADF with simulated critical values<br>BSADF date-stamping, minimum duration $\lceil \ln T \rceil$ ;<br>family-wise critical values for long monitoring |
| Volatility changes a lot (crypto, crises)                           | Wild bootstrap critical values                                                                                                                           |
| Is the rise accelerating super-exponentially?                       | LPPLS confidence indicator over many windows                                                                                                             |
| Has the risk state changed?                                         | Markov-switching filtered probabilities                                                                                                                  |
| How likely is a crash after a run-up?                               | Run-up conditioning across many assets (Greenwood, Shleifer and You 2019)                                                                                |

## Check Yourself, and Next: Chapter 18

### Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically?
- Why is the 95% critical value of GSADF far above that of a single ADF?
- Why is a single LPPLS critical time not a crash forecast?
- Why must hit rates after alarms be tested with a block bootstrap or rotations?

### : systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- CoVaR (the value-at-risk, VaR, of the system when an institution is in distress) and SRISK (the capital shortfall of an institution in a crisis) measure how much one institution adds to the risk of the system
- Further reading: Franke, Härdle & Hafner (2019), Kindleberger & Aliber (2005), Shiller (2015), Sornette (2017)

## Appendix: Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself  $h$  times:  $P_t = \sum_{i=1}^h \frac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \frac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$  (law of iterated expectations)
- Let  $h \rightarrow \infty$ :  $P_t = F_t + B_t$  with  $B_t = \lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition:**  $\lim_{h \rightarrow \infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$ ; it holds  $\Leftrightarrow B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period:  $\mathbb{E}_t[B_{t+1}] = \lim_h \mathbb{E}_t[P_{t+1+h}]/(1+r)^h = (1+r) B_t$ , the martingale condition of the rational bubble
- Seminar 17, Exercise A1 derives the general solution and why  $B_t \geq 0$  rules out a bubble that starts from zero [Diba & Grossman \(1988\)](#)

## Appendix: Bubbles for Fama: Effective Sample Size

- ▣ The run-ups fall in 31 calendar years and overlap: consecutive 2-year windows of the same boom share months
- ▣ Year-block bootstrap SE (9.1 percentage points) vs binomial SE (6.9): a design effect of 1.7, i.e. about 30 independent observations
- ▣ Blocks of 2, 3 and 5 consecutive years (2-year windows of adjacent years overlap):  $SE = 8.9, 8.7, 8.3$
- ▣ Unconditional mean 2-year return over all industry-months: 28%, above the CI of the mean return after a 100% run-up (−14% to 20%): returns after run-ups look lower than usual, but this is a descriptive comparison, not a formal test of the difference of two dependent estimates

## Appendix: Why the Full-Sample ADF Fails: the Leverage of Collapses

- ▣ OLS slope:  $\hat{b} = \sum_t \tilde{y}_{t-1} \Delta y_t / \sum_t \tilde{y}_{t-1}^2$ , with  $\tilde{y}_{t-1}$  the demeaned lagged log price
- ▣ Expansion:  $\Delta y_t > 0$  while  $\tilde{y}_{t-1}$  rises above its mean: positive terms
- ▣ Collapse: the most negative  $\Delta y_t$  occurs at the highest  $\tilde{y}_{t-1}$ : a few terms of the largest size and the opposite sign
- ▣ Evans form in logs,  $y = \ln B$ , collapse to  $\delta = e^c$  with probability  $1 - \pi$ ; for  $B \gg \delta$  the survival log growth is  $\approx \ln \frac{1+r}{\pi}$ :  
 $\mathbb{E}[\Delta y_t \mid y_{t-1}] \approx \pi \ln \frac{1+r}{\pi} + (1 - \pi)(c - y_{t-1})$ , slope  $-(1 - \pi) < 0$ : apparent mean reversion although  $\mathbb{E}_t B_{t+1} = (1 + r)B_t$   
 Evans (1991)
- ▣ Windows that contain the expansion but not the collapse have  $\hat{b} > 0$  when the log price accelerates there (next slide): a supremum over windows restores the power Phillips, Shi & Yu (2015b)

## Appendix: Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**:  $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$ , an autoregressive root  $1+r > 1$ 
  - ▶ In logs a pure bubble path  $B_t = B_0(1+r)^t$  has constant growth:  $\ln B_t = \ln B_0 + t \ln(1+r)$ , a deterministic linear trend in logs,  $b = 0$  in  $\Delta y_t = a + b y_{t-1} + \varepsilon_t$
- The log price  $\ln(F_t + B_t)$  **accelerates** while the bubble share grows: its growth rate rises from that of  $F$  towards  $r$  (a convex log path)
  - ▶ Once the bubble dominates, log growth is roughly constant again: the log-price regression then has little power
- So the log-price test targets a different alternative, **accelerating** log prices, not the level condition itself; the specification of the null and of the deterministic terms matters [Phillips, Shi & Yu \(2014\)](#)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq); PSY test the price-dividend ratio [Phillips, Wu & Yu \(2011\)](#), [Phillips, Shi & Yu \(2015a\)](#)

## Appendix: Limit Theory (1): the Null Distribution

- Null model:  $y_t = d T^{-\eta} + y_{t-1} + \varepsilon_t$ ,  $\eta > 1/2$ ,  $\varepsilon_t$  iid  $(0, \sigma^2)$ : a random walk with an asymptotically negligible drift [Phillips, Shi & Yu \(2015a\)](#)
- By the functional central limit theorem and continuous mapping (PSY, Theorem 1, eq. 5), with  $r_w = r_2 - r_1$  and  $W$  a standard Brownian motion:
  - ▶  $\text{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w [W(r_2)^2 - W(r_1)^2 - r_w] - \int_{r_1}^{r_2} W dr [W(r_2) - W(r_1)]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W^2 dr - \left( \int_{r_1}^{r_2} W dr \right)^2 \right\}^{1/2}}$
- $\sigma$  and  $d$  disappear: the limit is pivotal. SADF and GSADF are suprema of the same functional, so their critical values depend only on  $r_0$  and the deterministic terms
- $r_1 = 0$ ,  $r_2 = 1$  gives the Dickey–Fuller limit with a constant: its 95% quantile is negative (simulated:  $-0.06$ ), far from 1.645
- The drift and intercept specification changes the limit and the size [Phillips, Shi & Yu \(2014\)](#)

## Appendix: Limit Theory (2): Mildly Explosive Alternatives

- Alternative:  $y_t = \rho_T y_{t-1} + \varepsilon_t$  with  $\rho_T = 1 + c/k_T$ ,  $c > 0$ ,  $k_T \rightarrow \infty$ ,  $k_T = o(T)$ ; PSY use  $k_T = T^\alpha$ ,  $\alpha \in (0, 1)$ 
  - ▶ Closer to 1 than any fixed explosive root, farther than a local-to-unity root  $1 + c/T$ : the “mild” speed of real bubbles
- Phillips and Magdalinos:  $\frac{\rho_T^T}{\rho_T^2 - 1} (\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$ , a standard Cauchy law, without Gaussian errors [Phillips & Magdalinos \(2007\)](#)
  - ▶ Rate  $k_T \rho_T^T$ : exponentially fast; the ADF statistic diverges to  $+\infty$
- Hence SADF and GSADF diverge too (for GSADF any window of at least  $r_0 T$  observations inside the bubble suffices): the tests are consistent, power  $\rightarrow 1$  [Phillips, Shi & Yu \(2015b\)](#)
- Finite samples: power depends on  $c$ ,  $\alpha$  and the duration; short bursts may stay below the bar, as in the Evans simulation

## Appendix: Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8):  $\hat{r}_e = \inf\{r_2 \geq r_0 : \text{BSADF}_{r_2} > \text{scv}_{r_2}^{\beta_T}\}$ ,  
 $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \text{BSADF}_{r_2} < \text{scv}_{r_2}^{\beta_T}\}$
- $\text{scv}_{r_2}^{\beta_T}$ : the  $100(1 - \beta_T)\%$  quantile of BSADF on  $\lfloor Tr_2 \rfloor$  observations
  - Consistency,  $(\hat{r}_e, \hat{r}_f) \rightarrow_p (r_e, r_f)$ , needs  $\frac{1}{\text{scv}^{\beta_T}} + \frac{\text{scv}^{\beta_T}}{T^{1-\alpha/2}} \rightarrow 0$  (PSY, eq. 11)
- The critical value must diverge, so that the false-alarm probability vanishes ( $\beta_T \rightarrow 0$ ), but slower than  $T^{1-\alpha/2}$ , so that true bubbles are still found; e.g. growing like  $\ln T$  [Phillips, Shi & Yu \(2015a\)](#)
- In practice  $\beta_T$  is fixed at 5% at every date: each date keeps a 5% false-alarm chance, and over hundreds of dates a false alarm becomes likely (size study below)

## Appendix: The Price-Dividend Ratio: Comparison with PSY

- ▣ Our sample: 1 866 monthly levels, Jan 1871–Jun 2026; critical values from 1 000 Monte Carlo paths
- ▣ PSY, on data to 2010 and without our filter of 8 consecutive months, also date 1879–1880, 1955–1956 and 1986–1987 [Phillips, Shi & Yu \(2015a\)](#)
- ▣ With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too (Seminar 17, B1)

## Appendix: Family-Wise Critical Values: Bootstrap Design

- ▣ Phillips and Shi: the 95% quantile of the maximum of BSADF over  $\tau_b$  consecutive dates sets the FWER at 5% for the bootstrap design:  $\tau_b$  monitoring dates after a start-up of  $w_0$  observations [Phillips & Shi \(2020\)](#)
- ▣ Residuals of  $\Delta y_t = a + e_t$ , drawn with replacement and multiplied by  $N(0, 1)$  weights; paths of  $w_0 + \tau_b - 1$  changes; 999 replications
- ▣ Here  $\tau_b = 104$  weeks (two years); applied to BSADF over the full history, whose later dates search more start points, the 5% control is approximate, not proved
- ▣ The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

## Appendix: Real Time vs Hindsight: Definitions

- Start: first exceedance of the chain of BSADF episodes that leads to the peak, dated in hindsight
- Confirmed: after  $L$  consecutive exceedances, the first date the episode is known in real time, but with  $w_0$  and  $L$  set from the full-sample  $T$  (a truly real-time design fixes them ex ante)
- Leads in months (GameStop: calendar days converted to months); rise: from confirmation to the peak; fall: to the lowest close within 2 years (GameStop: 1 year)
- Bitcoin 2021: the explosive chain peaked at the local high of 13 Apr 2021 and ended in May 2021; the all-time high of 8 Nov 2021 (episodes table) came after a deep fall and a slower recovery that BSADF did not flag

## Appendix: Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate; Astill et al. compare monitoring statistics with the maximum over a training period, so the false-positive rate is set for a chosen monitoring horizon, robust to time-varying volatility [Astill et al. \(2018\)](#)
- Bubbles and crashes: after a bubble is detected, monitor for the crash with the sign of the mean of  $\Delta y_t$  (explosive vs stationary regime); never signals a crash before it starts [Whitehouse, Harvey & Leybourne \(2023\)](#)
- Earlier detection: alternative variance standardisations keep the false-detection probability of existing procedures but find bubbles sooner (applied to Bitcoin and housing) [Whitehouse, Harvey & Leybourne \(2025\)](#)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery [Phillips & Shi \(2018\)](#)
- Alternatives to sup-ADF: a Chow-type break test from random walk to explosive has the highest power in Homm and Breitung [Homm & Breitung \(2012\)](#); recursive tests with a start-point search [Harvey, Leybourne & Sollis \(2015\)](#)

## Appendix: LPPL in the Cryptocurrency and Romanian Markets

- ▣ Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash [Pele, Mazurencu & Nijkamp \(2013\)](#); crash modelling for the BVB [Pele & Mazurencu-Marinescu \(2012\)](#)
- ▣ Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles [Pele & Mazurencu-Marinescu-Pele \(2019\)](#), [Conda et al. \(2026\)](#)
- ▣ Dogecoin 2021: LPPL together with sentiment from social media [Pele \(2025\)](#)
- ▣ Common lesson: LPPL describes the acceleration well; the crash date remains the weak point



## Appendix: Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood by prediction and update:  $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1}) \varphi(r_t; \mu_j, \sigma_j^2)$ ,  $\mathcal{F}_t$  = returns up to  $t$ ,  $\varphi(x; \mu, \sigma^2)$  the Normal density with mean  $\mu$  and variance  $\sigma^2$  [Hamilton \(1989\)](#)
- Kim smoother, backwards:  $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \frac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$  [Kim \(1994\)](#)
- EM algorithm: with smoothed  $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$  and  $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$ , the M-step is weighted least squares [Hamilton \(1990\)](#)
  - ▶  $\hat{\mu}_j = \frac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}$ ,  $\hat{\sigma}_j^2 = \frac{\sum_t \xi_{tj} (r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}$ ,  $\hat{p}_{ij} = \frac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- The likelihood is unbounded (one regime on a single return with  $\sigma_j \rightarrow 0$ ) and has several local maxima (regime labels can switch): we take the best interior maximum from 20 random starts and check that no  $\hat{\sigma}_j$  collapses towards 0, also in bootstrap refits
- Standard errors from the inverse numerical Hessian; they assume the number of regimes is known

## Appendix: Why the LR Test for Regimes Is Non-Standard

- Under  $H_0$  the regime means and variances coincide, so  $p_{11}, p_{22}$  are not identified (nuisance parameters present only under  $H_1$ ); the information matrix is singular and some scores are identically zero [Hansen \(1992\)](#)
- Hansen: bounds on the standardised LR via empirical-process theory [Hansen \(1992\)](#)
- Carrasco, Hu and Ploberger: an optimal test needing only the estimate under  $H_0$ , with bootstrap critical values [Carrasco, Hu & Ploberger \(2014\)](#)
- Same logic for two vs three regimes: the null model is then the fitted two-regime chain

## Appendix: Case Study, Data and Method

- Data (Section 1.1): 51 580 408 arm's-length home sales from deeds records, 1995–2014, in 115 metropolitan areas (48% of the US housing stock); the cycle analysis covers 2000–2011, since realised holding periods need consecutive sales
  - ▶ Listings of homes for sale linked to the sales at property level in 57 of these areas
- Speculators (Section 1.2): short-term buyers resell within three years (realised holding period); non-occupant buyers give a mailing address different from the property address (102 areas)
- Lead-lag (Eq. 1, Figure 2):  $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$  for  $k = -12, \dots, 48$  months
  - ▶  $p$ : log price, demeaned by city;  $v$ : sales divided by the 2000 housing stock, demeaned by city and calendar month; implied correlation  $\beta_k \text{sd}(v_{i,t-k}) / \text{sd}(p_{i,t})$ , standard errors clustered by month
- Cross-section (Table 2): OLS across cities of the price boom (2000–2006), price bust (2006–2011), listings, volume and foreclosures on the speculative volume boom (2000–2005)
- Model (Sections 5–6): buyers extrapolate recent price growth and differ in horizon; sellers post list prices and not every listing sells (non-Walrasian market)
  - ▶ Counterfactuals: rational expectations, no speculators, transaction taxes

## Appendix: Case Study, Complete Results

- Figure 2: the correlation between prices and lagged volume peaks at a lag of 24 months; Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005), short-term sales from 510 to 940 thousand, 39% of the rise (non-occupant purchases: 53%)
  - ▶ Interpretation: volume leads prices by about two years, and speculators account for much of the volume boom
- Table 2, Panel A: coefficient of the short-term sales volume boom in the price-boom regression 1.930 (standard error 0.297,  $R^2 = 0.272$ ); one standard deviation more (12.9%, Table 1) means a price rise larger by 24.9 percentage points; price bust  $-0.571$  (0.083), 7.4 points deeper
  - ▶ Interpretation: cities with more speculation had both larger booms and larger busts
- Table 2, Panels B–C (quantities in % of 2003 total volume): unsold listings in the quiet 5.961 (1.353), 76.9 points more per standard deviation; foreclosures 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas (Section 3.2)
  - ▶ Interpretation: consistent with speculators who bought in the boom listing in the quiet and failing to sell; that this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year goes with an 8.2 points larger short-term buyer share; Table 6: in the model, a 5% purchase tax on all buyers cuts the price bust from 8.2% to 1.1%, on non-occupants only to 5.8%
  - ▶ Interpretation: extrapolation draws short-horizon buyers in; short horizons, not non-occupancy, amplify the cycle

## Appendix: Case Study, Lead–Lag Inference

- $p$  and  $v$  demeaned over the full sample; implied correlation  $\hat{\beta}_k \text{sd}(v_{t-k})/\text{sd}(p_t)$ ,  $k = -12, \dots, 48$  months
- At  $k = 0$  the correlation is 0.03, at  $k = -12$  it is  $-0.34$ : level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean: the “implied correlation” is a standardised slope, not a Pearson correlation, so bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series, doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory
- 1990–2011: the maximum moves to the edge of the grid,  $k = 48$  (0.82)

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** map what is known, then verify every reference
  - ▶ “Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design.”
  - ▶ Resolve each DOI on [Crossref](#) and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing explanations, each with its own observable
  - ▶ “For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts.”
  - ▶ E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code:** on the course data, replicating a known number first
  - ▶ “Implement GSADF with  $r_0 = 0.01 + 1.8/\sqrt{T}$  on weekly Bitcoin closes from the course data and simulate its 95% critical value.”
  - ▶ Replicate first:  $GSADF = 3.85$  and critical value 2.27 on 626 weeks, as in this chapter

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
  - ▶ “Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.”
  - ▶ Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
  - ▶ “Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?”
  - ▶ Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid  $p$ -values, all 12 cells, the alarms that failed
  - ▶ The prompts that shaped the design go into AI\_USE.md

## Appendix: What the Researcher Must Check

- **Full-sample information in date-stamping:** critical values, lag order or detrending fitted on the whole sample
  - ▶ Test: cut the sample at  $t$  and recompute; the flag at  $t$  must not change
- **Wrong tail:** left-tail ADF critical values (about  $-2.86$ ) used against an explosive alternative
  - ▶ Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility:** Monte Carlo critical values under constant variance distort the size
  - ▶ Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity:** many dates, assets and windows; some alarm is almost certain
  - ▶ Test: run the monitoring rule on simulated random walks and count the paths with at least one alarm
- **Overlapping outcomes:** 90-day windows overlap, so Fisher's exact test or a  $\chi^2$  test overstates significance; windows cut by the end of the data must be dropped
  - ▶ Test: block bootstrap or circular rotations of the alarm series, which keep the dependence of each series; rotations also need stationarity (mini-case)

## Appendix: Mini-Case: What the Result Does and Does Not Say

- A large one-sided  $p$  is not evidence of **no** information: it only fails to support a **higher** crash probability after alarms
  - ▶ Both point estimates are **negative**: rotation  $p$  for a lower crash rate after alarms 0.07 (Bitcoin), 0.03 (Nasdaq 100); block-bootstrap CI and Newey–West test in Seminar 17, B6
  - ▶ A post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days, not an announced crash
- Assumptions behind the rotation  $p$ -values: under  $H_0$  every circular shift of the alarm series (at least 26 steps, both ends joined) is as likely as the observed alignment
  - ▶ This is stronger than equal hit rates: it needs stationary series and alarms independent of the crash process; shared regimes or a time-varying crash rate can invalidate it
  - ▶ Hence the  $p$ -values are exploratory; their size should be checked by simulation under plausible null processes
- Outcomes: only dates whose whole horizon is observed (no censored windows at the end of the data)

## Appendix: Further Study on Quantinar

### Advanced courses on quantinar.com

- ☐  Cryptocurrency: Speculative Asset and Medium of Exchange
- ☐  Are Cryptos becoming alternative Assets?
- ☐  Statistics of Financial Markets

### Code and data

- ☐ Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_17](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_17)

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